CHAPTER

20 Quantum Mechanics

20.1 INTRODUCTION

In 1925, de Broglie introduced the concept of matter waves and the idea of wave-particle duality. He suggested that the wave-particle duality observed in case of light should be extended to microparticles also. The combination of the idea of quantization with the idea of wave-particle duality proved to be very fruitful for the development of quantum mechanics. The whole apparatus of quantum mechanics was built in 1925-26. In 1925 Heisenberg suggested that any reference to conceptual pictures which are not amenable to direct experimental verification, should be discarded. He formulated matrix mechanics which is set in terms of the observable quantities alone. In 1926, Schrödinger developed wave mechanics. His theory is based on explicit use of a mental picture of matter wave which replaced the classical picture of point particle. He developed the well-known differential equation for a wave function. The problem of calculating the energy levels of a bound microparticle was reduced by Schrödinger to the problem of finding eigenvalues. Heisenberg's theory came to be known as **matrix mechanics** while that of Schrödinger as **wave mechanics**. In 1926, Max Born proposed the probability interpretation of the wave function. In 1927 the wave behaviour of microparticles was confirmed by experiments on electron diffraction conducted simultaneously in several different laboratories. In 1927, Heisenberg introduced the uncertainty principle. Through this principle Heisenberg showed how the concepts of coordinate, momentum, energy etc should be applied to microparticles. The uncertainty principle marked the final break of quantum mechanics from classical determinism and established quantum mechanics as a statistical theory. In quantum mechanics, the waves and particles are not classes of objects. They are distinct modes of behaviour shared by all atomic particles. Every microparticle can behave like a particle and like a wave too. In 1930, P.A.M.Dirac proposed a general formalism which is a unifying concept of the matrix mechanics and wave mechanics.

The new laws applicable for atoms and subatomic particles constitute **quantum mechanics**. The laws of conservation of momentum, angular momentum and energy are still valid but we are not in a position to obtain on their basis a detailed description of the motion of the subatomic particles. New ideas such as quantization of physical quantities and allowed values of physical quantities are required to be incorporated into the theory. Planck's hypothesis of energy quanta, Einstein's ideas on photons, de Broglie's visualization of the wave properties of micro-particles and the Heisenberg's uncertainty principle provided the basis for the development of quantum mechanics.

20.2 DE BROGLIE HYPOTHESIS

In 1924, Louis de Broglie extended the wave–particle dualism of light to the material particles. He reasoned out that nature exhibits a great amount of symmetry. Therefore, if a light wave can act as a wave sometimes and as a particle at other times, then particles such as electrons should also act as waves at times. This is known as *de Broglie hypothesis*.

According to de Broglie hypothesis any moving particle is associated with a wave. The waves associated with particles are known as **de Broglie waves** or **matter waves**. The wavelength λ of matter waves associated with a particle moving with velocity υ is inversely proportional to the magnitude of the momentum of the particle. Thus,

$$\lambda = \frac{h}{m\upsilon} \tag{20.1}$$

De Broglie deduced the connection between the particle and wave properties as follows.

De Broglie wavelength of Matter waves

As a photon travels with the velocity c, we can express its momentum as

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$$

Thus, the wavelength and momentum p of a photon are related to each other through the expression

$$\lambda = \frac{h}{p} \tag{20.2}$$

De Broglie proposed that the relation (20.2) between the momentum and the wavelength of a photon is a universal one and must be applicable to photons and material particles as well.

The quantities ν and λ are wave properties and the quantities E and p are particle properties. They are tied to each other through the relations

$$E = hv \text{ and } p = h/\lambda$$

which demonstrate that the wave and particle natures of a photon are intimately tied up to each other. These equations reflect the wave-particle dualism of light.

Now let us consider a moving particle. A particle of mass 'm' moving with a velocity υ carries a momentum $p = mv$ and it must be associated with a wave of wavelength

$$\lambda = \frac{h}{p} = \frac{h}{m\upsilon} \tag{20.3}$$

The waves associated with moving particles are called **matter waves** or **de Broglie waves.**

The relation $\lambda = h/m\upsilon$ is known as **de Broglie equation** and the wavelength λ is called the **de Broglie wavelength**.

From equ.(20.3), we may draw the following conclusions.

1. $\lambda \to \infty$ when the velocity of the particle is zero. It means that matter waves are detectable only for moving particles.
2. Lighter the particle, smaller the value of mass m and hence the longer is the wavelength of the matter wave associated with it. Therefore, wave behaviour of micro-particles will be significant whereas waves associated with macro-bodies can never be detected.

3. The smaller the velocity of the micro-particle, the longer is the wavelength of the matter wave associated with it.

If a photon is considered to be a particle, then the corresponding electromagnetic wave is the de Broglie wave for the photon. Similarly, atomic particles also can be viewed as associated with matter waves, which do not have any similarity to any known waves. It is later understood that the waves associated with particles are not real three dimensional waves in the way sound waves are, but are **probability waves** related to the probabilities of finding the particles in various places and with various properties.

De-Broglie wavelength associated with an accelerated charged particle

If a charged particle, say an electron is accelerated by a potential difference of V volts, then its kinetic energy is given by $K.E. = eV$.

Or $$\frac{1}{2}m\upsilon^2 = eV$$

$\therefore$ $$\upsilon = \sqrt{\frac{2eV}{m}}$$

Then the electron wavelength is given by

$$\lambda = \frac{h}{m\upsilon} = \frac{h}{m}\cdot\sqrt{\frac{m}{2eV}}.$$

$\therefore$ $$\lambda = \frac{h}{\sqrt{2emV}} \tag{20.4}$$

De Broglie Wavelength expressed in terms of K.E.

If a particle has kinetic energy K.E., then $\text{K.E.} = \frac{1}{2}m\upsilon^2 = \frac{m^2\upsilon^2}{2m} = \frac{p^2}{2m}$

or $$p = \sqrt{2m(K.E.)}$$

$\therefore$ $$\lambda = \frac{h}{\sqrt{2m(K.E.)}} \tag{20.5}$$

De Broglie wavelength associated with particles in thermal equilibrium

If particles are in thermal equilibrium at temperature T, then their kinetic energy is given by

$$\text{K.E.} = \frac{3}{2}kT$$

$\therefore$ $$\lambda = \frac{h}{\sqrt{2m(K.E.)}} = \frac{h}{\sqrt{3mkT}} \tag{20.6}$$

20.3 DE BROGLIE'S JUSTIFICATION OF BOHR'S POSTULATE

Bohr's theory of atomic structure was successful in explaining a large body of experimental observations concerning atomic behaviour but the three adhoc postulates underlying his theory remained without a valid theoretical justification for a long time. In support of his hypothesis of matter waves, de Broglie demonstrated that it could provide an explanation for the postulate regarding quantization of angular momentum of electron in Bohr's model of atom.

One of the postulates that Bohr used in formulating a model of atom is that the angular momentum L of the electron revolving in a stationary orbit is quantized. Thus,

$$L = n\hbar \tag{20.7}$$

The above postulate of Bohr follows directly from the concept of matter waves.

As the electron travels round in one of its circular orbits, the associated matter waves propagate along the circumference again and again. A wave must meet itself after going round one full circumference. If it did not meet, the wave would be out of phase with itself after going round one orbit. After a large number of orbits, all possible phases would be obtained and the wave would be annihilated by destructive interference. It implies that the wave should produce a standing wave profile in the orbit to preclude the electron energy from radiating away.

If a stretched string is fastened at both ends and is made to vibrate, standing waves are formed provided the length of the string is an integral number of half–wavelengths of the disturbance. If the string is formed into a circular loop, the condition for standing waves is that the circumference of the loop should be an integral number of whole wavelengths of the disturbance. Thus, if r is the radius of the circular loop,

$$2\pi r = n\lambda \qquad n = (1,2,3,....) \tag{20.8}$$

We may regard the stationary electron orbits in an atom to be analogous to the circular loop of string. We conclude that stationary electron wave pattern can form in the orbit if only an integral number of electron wavelengths fit into the orbit, as shown in Fig. 20.1 (*b*).

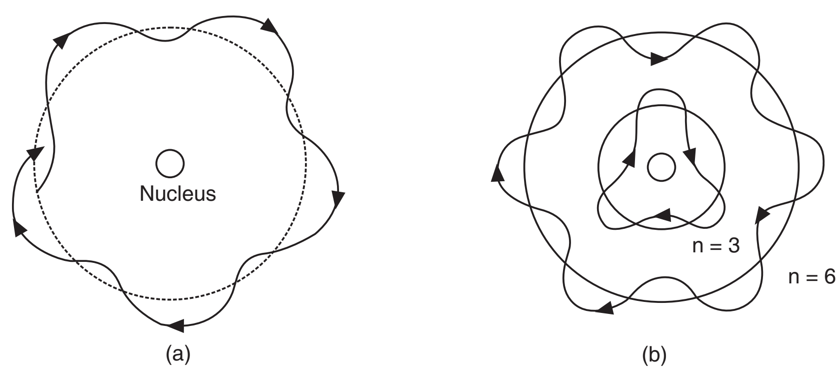

Fig. 20.1

The above equation (20.8) can be applied for electron waves, taking λ as de Broglie wavelength of electron waves.

The de Broglie wavelength of electron wave is given by

$$\lambda = \frac{h}{m\upsilon}$$

where υ is the speed of the electron in the orbit. Using the de Broglie wavelength into equ.(20.8), we obtain

$$2\pi r = \frac{nh}{m\upsilon}$$

$$\therefore \qquad m\upsilon r = \frac{nh}{2\pi} \tag{20.9}$$

But the quantity '$m\upsilon r$' is the angular momentum, L, of electron in the orbit of radius r.

Thus, $$L = m\upsilon r$$

(*i*) When the atomic particle velocity is *non-relativistic*, the total energy $E = mc^2$ and momentum $p = m\upsilon$.

Therefore, the phase velocity of the de Broglie wave associated with the particle is

$$\upsilon_p = \frac{E}{p} = \frac{mc^2}{m\upsilon} = \frac{c^2}{\upsilon} \tag{20.13}$$

As $\upsilon < c$, the phase velocity of the de Broglie wave associated with the atomic particle is always greater than c.

(*ii*) When the atomic particle velocity is *relativistic*, the total energy $E = \sqrt{m_o^2c^4 + p^2c^2}$, where m_o is the rest mass of the particle.

Therefore, the phase velocity of the de Broglie wave associated with the particle is

$$\upsilon_p = \frac{E}{p} = \left[\frac{m_o^2c^4 + p^2c^2}{p^2}\right]^{1/2}$$

$$= c\left[\frac{m_o^2c^2}{p^2} + 1\right]^{1/2} = c\left[\frac{m_o^2c^2\lambda^2}{h^2} + 1\right]^{1/2} \tag{20.14}$$

As the term $\frac{m_o^2c^2\lambda^2}{h^2}$ is always a positive quantity, the phase velocity of the de Broglie wave associated with the atomic particle is always greater than c.

According to the theory of relativity, it is not possible that the velocity of the particle wave be greater than or equal to the velocity of light. Hence, a harmonic wave of wavelength λ cannot represent a moving atomic particle. Thus, ***de Broglie waves cannot be harmonic waves***.

20.9 WAVE PACKET – REPRESENTS A MICROPARTICLE

We have so far assumed that a particle may be represented by a monochromatic de Broglie wave. However, a wave spreads over a large region of space and cannot represent a highly localized particle. Schrödinger postulated that a **wave packet** rather than a single harmonic wave represents a particle. A wave packet consists of a group of harmonic waves. Each wave has slightly different wavelength. The superposition of a very large number of harmonic waves differing infinitesimally in frequency will produce a single wave packet (see Fig. 20.6 c). The waves interfere constructively over only a small region of space and cancel each other everywhere except in that small region. The position of the particle would then be approximately determined by the position of the wave packet.

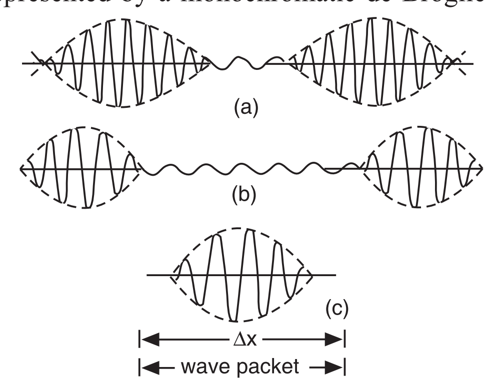

Fig. 20.6: Formation of a Wave packet. (a) two waves of slightly different frequencies produce constructive interference. (b) three waves produce interference maxima of larger size separated by larger distance. (c) A large number of waves-having slightly different frequencies produces only one maxima and it is called a wave packet.

The velocity with which the wave packet propagates is called the **group velocity** υ_g.

Example 20.2. *An enclosure filled with helium is heated to 400K. A beam of He-atoms emerges out of the enclosure. Calculate the de Broglie wavelength corresponding to He atoms. Mass of He atom is 6.7×10^{-27}kg.*

Solution. De Broglie wavelength $\lambda = \dfrac{h}{\sqrt{2mkT}}$

$$= \frac{6.63 \times 10^{-34}\, Js}{\sqrt{2 \times 6.7 \times 10^{-27}\, kg \times 1.376 \times 10^{-21}\, J/\deg \times 400}}$$

$$= \mathbf{0.769\ \text{Å}}$$

Example 20.3: *Find the de Broglie wavelength of*

(*i*) *an electron accelerated through a potential difference of 182 volts, and*

(*ii*) *a 1 kg object moving with a speed 1 m/s. Comparing the results explain why the wave nature of matter is not more apparent in daily observations.*

Solution:

(*i*) $$\lambda_e = \frac{h}{\sqrt{2emV}} = \frac{6.626 \times 10^{-34}\, J.s}{\sqrt{2\left(1.602 \times 10^{-19}\, C\right)\left(9.11 \times 10^{-31}\, kg\right)182V}}$$

$$= \frac{6.626 \times 10^{-34}\, J.s.}{7.29 \times 10^{-24}\, kg.m/s} = 0.91 \times 10^{-10} \frac{kg.m^2/s}{kg.m/s} = 9.1 \times 10^{-11}\, m = \mathbf{0.91\,\text{Å}}$$

(*ii*) $$\lambda_m = \frac{h}{M\upsilon} = \frac{6.626 \times 10^{-34}\, J.s}{1kg \times 1m/s} = 6.6 \times 10^{-34} \frac{kg.m^2/s}{kg.m/s} = \mathbf{6.6 \times 10^{-34}\ m}\ .$$

It is seen from the above that the wavelength of the accelerated electron is about 10^5 times larger then its own size ($\approx 10^{-15}$ m) and is therefore significant. On the other hand, the wavelength associated with the macroscopic object is negligibly small and is thus not apparent in its interactions with other objects.

20.11 HEISENBERG UNCERTAINTY PRINCIPLE

The wave nature of atomic particles leads to some inevitable consequences. Classically, the state of a particle can be defined by specifying its position and momentum at any given time t. If a body is moving along x-direction with a velocity υ, its position is given by $x = \upsilon\, t$ and its momentum by $p = m\upsilon$. From this,

$$x = \frac{p}{m}t \tag{20.28}$$

At each instant, the position and momentum can be measured to a very high accuracy. When an atomic particle is conceptualized as a de Broglie wave packet such a precision becomes restricted.

Schrödinger postulated that a moving microparticle is equivalent to a wave packet. A wave packet spreads over a region of space. Therefore, it is difficult to locate the exact position of the microparticle. Although the particle is somewhere within the wave packet, it is impossible to know where exactly the particle is at a given instant. If the linear spread of the wave packet is Δx, the particle would be located somewhere within the region Δx. The probability of finding the particle is a maximum at the centre of the wave packet and falls off to zero at its ends. Therefore, there is an **uncertainty** Δx in the position of the particle. As a result, the momentum of the particle at that instant cannot be determined precisely. It means that the location and momentum of a microparticle cannot be **simultaneously** determined with certainty. Any attempt to determine these variables will lead to uncertainties in each of the variables.

In 1927 Heisenberg showed that the product of uncertainty Δx in the x-coordinate of a quantum particle and the uncertainty Δp_x in the x-component of the momentum would always be of the order of Planck's constant h. Thus,

$$\Delta x \cdot \Delta p_x \approx h$$

or more precisely

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} \tag{20.29}$$

This is known as Heisenberg's uncertainty principle for position and momentum, which may be stated as follows:

"*It is not possible to know simultaneously and with exactness both the position and the momentum of a microparticle*".

The Uncertainty Principle implies a built-in, unavoidable *limit to the accuracy with which we can make measurements*. Classically, it is thought that the precision of any measurement was limited only by the accuracy of the instruments the experimenter used. Heisenberg showed that whatever may be the accuracy of the instruments used, quantum mechanics limits the precision when two properties are measured at the same time. These are *not just any* two properties but pairs of measurable quantities whose product has dimensions of energy × time. Such quantities are called *conjugate quantities* in quantum mechanics, and have a special relation to each other. Position–linear momentum, energy-time, time-frequency and angular momentum-angular displacement are conjugate pairs of variables.

The uncertainty principle asserts that it is physically impossible to know simultaneously the exact position ($\Delta x = 0$) and exact momentum ($\Delta p_x = 0$) of a micro-particle. According to it, the more precisely we know the position of the particle, the less precise is our information about its momentum. To localize a wave packet, we have to add more wavelengths to form the wave packet. More wavelengths mean larger $\Delta\lambda$ and more uncertainty in momentum (note that $\Delta p \propto \Delta\lambda$). Conversely, in order to have more precise value of momentum, the wave packet should contain less number of waves. Less number of waves produces a longer wave packet. Thus, the momentum of a particle cannot be precisely specified without our loss of knowledge of the position of the particle at that time. Similarly, a particle cannot be precisely localized in a particular direction without our loss of knowledge of momentum in that particular direction. We can at best specify that certain momentum of the particle is more probable than the other or that the particle is more likely to be here than there. We cannot use classical notions like coordinates and momentum to describe the motion of quantum particles. Thus, the uncertainty principle implies that we can never define the path of an atomic particle with the absolute precision indicated in classical mechanics. Therefore, concepts such as velocity, position, and acceleration are of limited use in quantum world.

Relations similar to (20.29) hold good for other components of position and linear momentum. Thus,

$$\Delta y \cdot \Delta p_y \geq \frac{\hbar}{2} \tag{20.29a}$$

$$\Delta z \cdot \Delta p_z \geq \frac{\hbar}{2} \tag{20.29b}$$

20.11.1 Energy - Momentum Uncertainty

The uncertainty relation for the simultaneous measurement of energy *E* and time *t* is expressed as

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \tag{20.30}$$

The physical significance of the energy-time uncertainty relation is different from that of the position –momentum uncertainty. If ΔE is the maximum uncertainty in the determination of the energy of a particle, then the minimum time interval for which the particle remains in that state is given by

$$\Delta t = \frac{\hbar / 2}{\Delta E}$$

And, if a particle remains in a particular energy state for a maximum time Δt, then the minimum uncertainty in the particle energy is given by

$$\Delta E = \frac{\hbar / 2}{\Delta t}$$

Derivation: We can obtain the result (20.30) as follows. Let us consider a microparticle of mass *m* moving with a velocity υ. Its kinetic energy will be

$$E = \frac{1}{2} m \upsilon^2$$

If the uncertainty in the energy is ΔE, then $\Delta E = \Delta\left[\frac{1}{2} m \upsilon^2\right] = m\upsilon\, \Delta\upsilon = \upsilon\, \Delta p$

As the velocity $\upsilon = \frac{\Delta x}{\Delta t}$, the uncertainty in energy may be written as $\Delta E = \frac{\Delta x}{\Delta t} \Delta p$

Thus, $\Delta E \cdot \Delta t = \Delta x \cdot \Delta p$

But $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$

Therefore, $\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$

The above relations are to be supplemented by the following uncertainty relation

$$\Delta M_x \cdot \Delta \varphi_x \geq \frac{\hbar}{2} \tag{20.31}$$

where ΔM_x is the uncertainty in the projection of the angular momentum on the x-axis and $\Delta \Phi_x$ is the uncertainty in the angular coordinates of the microparticle.

By analogy with (20.28a) and (20.28b), we may write down relations for other projections of momentum and angular momentum as follows:

$$\Delta M_x \cdot \Delta \varphi_y \geq \hbar/2 \text{ and } \Delta M_z \cdot \Delta \varphi_z \geq \hbar/2 \tag{20.31a}$$

In general if *q* and *p* denote two canonically conjugate variables, the uncertainty relation is given by

$$\Delta q \cdot \Delta p \geq \hbar/2 \tag{20.32}$$

The above relations do not mean that the uncertainty principle creates certain obstacles to the understanding of the atomic phenomena; it only reflects certain peculiarities of the objective properties of a quantum particle.

20.12 ELEMENTARY PROOF OF UNCERTAINTY PRINCIPLE USING DE BROGLIE WAVE CONCEPT

A wave packet produced by a superposition of large number of harmonic waves is shown in Fig. 20.10. Since a wave packet is not an infinite harmonic wave, it has a range of wave numbers Δk instead of one definite wave number. Δk is thus the uncertainty in wave number. Further, the position of the particle cannot be given with certainty. It will lie somewhere between the two consecutive nodes (see Fig. 20.7b and Fig. 20.10). Thus, the uncertainty in the position of the particle is equal to the distance between two consecutive nodes. Referring to equ. (20.15), the condition for formation of a node is

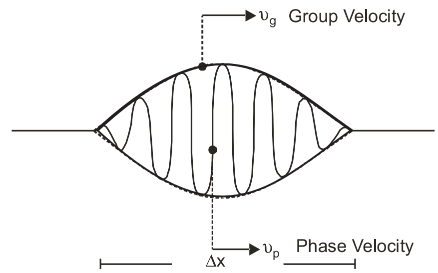

Fig. 20.10

$$\cos\left(\frac{\Delta\omega t}{2}-\frac{\Delta kx}{2}\right) = 0$$

i.e.,
$$\frac{\Delta\omega}{2}t-\frac{\Delta k}{2}x = \frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2},\ldots,(2n+1)\frac{\pi}{2}$$

Thus, if x_1 and x_2 are the positions of two consecutive nodes, then

$$\frac{\Delta\omega}{2}t-\frac{\Delta k}{2}x_1 = (2n+1)\frac{\pi}{2}$$

and
$$\frac{\Delta\omega}{2}t-\frac{\Delta k}{2}x_2 = (2n+3)\frac{\pi}{2}$$

Subtracting the above upper equation from the lower one, we find that

$$\frac{\Delta k}{2}(x_2-x_1) = \pi \quad \text{or} \quad \frac{\Delta k}{2}\Delta x=\pi$$

where $\Delta x = (x_2 - x_1)$ is the uncertainty in the position of the particle. Thus,

$$\Delta x = \frac{2\pi}{\Delta k}.$$

Since
$$k = \frac{2\pi}{\lambda}=\frac{2\pi p}{h},$$

$$\Delta k = \frac{2\pi}{h}\Delta p$$

Therefore,
$$\Delta x = \frac{2\pi}{\Delta k}=\frac{2\pi h}{2\pi\,\Delta p}=\frac{h}{\Delta p}$$

or
$$\Delta x\cdot\Delta p = h$$

When we consider a group consisting of very large number of harmonic waves of continuously varying frequencies, the product of the uncertainties comes to

$$\Delta x\cdot\Delta p \geq \frac{1}{2}\hbar$$

20.13 IMPLICATION OF UNCERTAINTY PRINCIPLE

The uncertainty principle expresses a fundamental limitation in nature that also limits the precision of our measurements. According to classical mechanics the position and momentum of a macro-particle can be determined exactly. But the uncertainty principle asserts that it is physically impossible to know simultaneously the exact position ($\Delta x = 0$) and exact momentum ($\Delta p_x = 0$) of a microparticle. According to it, the more precisely we know the posistion of the particle, the less precise is our information about its momentum. Thus, the momentum of a particle cannot be precisely specified without our loss of knowledge of the position of the particle at that time. Similarly, a particle cannot be precisely localized in a particular direction without our loss of knowledge of momentum in that particular direction. We can at best specify that certain momentum of the particle is more probable than the other or that the particle is more likely to be here than there. It means that our classical notions like coordinates and momentum derived from ordinary macroscopic experiences are inadequate to describe the atomic world. The uncertainty principle points out that in the microscopic world,

(1) the dynamical variables of a particle are combined in sets of *simultaneously determined* quantities which are known as *complete sets* of quantities;
(2) the coordinate and momentum components of a particle etc are **pairs of concepts** which are interrelated and fall in different complete sets of quantities. They cannot be defined simultaneously in a precise way.

Thus, the uncertainty principle implies that we can never define the path of an atomic particle with the absolute precision indicated in classical mechanics. Therefore, concepts such as velocity, position, and acceleration are of limited use in quantum world. To describe the quantum particle the concept of *energy* becomes important since it is related to the *state* of the system rather than to its path.

20.14 UNCERTAINTY PRINCIPLE IS NOT SIGNIFICANT IN CASE OF MACRO-BODIES

The Heisenberg Principle is of no practical importance for heavy bodies where the de Broglie wavelength is negligibly small.

For example, let us take the case of a cricket ball in flight. The indeterminacy in the position of the ball is, say, 1 mm. We can determine the indeterminacy of velocity of the ball from uncertainty principle.

$$\Delta x \cdot \Delta p \approx h$$

$$\therefore \qquad \Delta x \cdot m\Delta \upsilon \approx h$$

$$\Delta \upsilon \approx \frac{h}{m\Delta x} = \frac{6.62\times10^{-34}\,J.s}{0.5kg\times10^{-3}\,m} \cong 10^{-30}\,m/s.$$

The above inaccuracy is negligible and not detectable. It implies that the uncertainties are of no importance in case of macro bodies; and the position and velocity of a macro body can be simultaneously determined with a high degree of accuracy. As a result, macroscopic body follows a well defined trajectory.

In contrast if we take the example of an electron orbiting in a hydrogen atom, the inaccuracy in its position is ± 1Å. The uncertainty in its speed is

$$\Delta \upsilon = \frac{h}{m\Delta x} = \frac{6.62\times10^{-34}\,J.s}{9.11\times10^{-31}\,kg\times2\times10^{-10}\,m} \approx 2\times10^{5}\,m/s$$

which is of the same order as the velocity of the electron in the orbit. It means that it is not possible to determine the velocity and the position of a microparticle with certainty and as such we cannot talk of a specific trajectory. Instead we have to be content knowing only the probable values.

is about 10^5 times smaller than the wavelength of light. Therefore, we use a γ-ray microscope to detect and locate an electron.

Let a free electron be directly beneath the center of the γ-ray microscope's lens. The circular lens forms a cone of angle 2α from the electron. The electron is illuminated from the left by γ-rays. The microscope can resolve objects to a size of Δx. Δx is given by the expression

$$\Delta x = \frac{\lambda}{2\sin\alpha} \qquad (20.35)$$

To be observed by the microscope, the γ-ray must be scattered into any angle within the cone of angle 2α. A γ-photon carries a very large momentum. When the γ-photon strikes the electron, part of the momentum and energy are transferred to the electron due to Compton scattering. Consequently, as the scattered photon enters the microscope, the electron has already moved away in a certain direction (Fig. 20.12). The total momentum p is related to the wavelength by the formula $p = \frac{h}{\lambda}$.

In the extreme case of diffraction of the gamma ray to the right edge of the lens, the total momentum in the x direction would be the sum of the electron's momentum p'_x in the X direction and the gamma ray's momentum in the x-direction:

$$p'_x + \left(\frac{h\ \sin\alpha}{\lambda'}\right) \qquad (20.36)$$

where λ′ is the wavelength of the deflected gamma ray. In the other extreme, the observed gamma ray recoils backward, just hitting the left edge of the lens. In this case, the total momentum in the X-direction is:

$$p''_x - \left(\frac{h\ \sin\alpha}{\lambda''}\right) \qquad (20.37)$$

The final X-momentum in each case must equal the initial X-momentum, since momentum is *conserved*. Therefore, the final X-momenta are equal to each other:

$$p'_x + \left(\frac{h\ \sin\alpha}{\lambda'}\right) = p''_x - \left(\frac{h\ \sin\alpha}{\lambda''}\right)$$

If α is small, then the wavelengths are approximately the same, $\lambda' \approx \lambda'' \approx \lambda$. And we have

$$p''_x - p'_x = \frac{2h\ \sin\alpha}{\lambda}$$

or

$$\Delta p_x = \frac{2h\ \sin\alpha}{\lambda} \qquad (20.38)$$

Using eq.(20.35) into the above equation, we obtain

$$\Delta p_x = \frac{h}{\Delta x}$$

$$\therefore \qquad \Delta x \cdot \Delta p_x = h$$

20.16 APPLICATIONS OF UNCERTAINTY PRINCIPLE

We deal here with three simple examples to illustrate the application of uncertainty principle.

(a) Bohr's Orbit and Energy

Let us consider the electron in a hydrogen atom. We cannot know at any instant the position of the electron in its orbit. It might be on the left or right of the nucleus, as sketched in

Fig. 20.13. The electron position has an uncertainty $\pm r$. We cannot know likewise whether the electron is moving upward or downward .The uncertainty in its velocity therefore $\pm \upsilon$. Taking $\Delta x = r \approx 0.5 \times 10^{-10}$ m, the uncertainty in the electron speed is

$$\Delta \upsilon = \frac{h}{2\pi m \Delta x}$$

$$= \frac{6.62 \times 10^{-34} J.s}{2 \times 3.124 \times 9.11 \times 10^{-31} kg \times 0.5 \times 10^{-10} \text{m}}$$

$$\cong 2 \times 10^{6} \text{ m/s}$$

The velocity 'υ', of an electron in an atom is of the order of 1.0×10^{6} m/s and is of the same order as the uncertainty $\Delta \upsilon$. Therefore, we conclude that the uncertainty in momentum is of the same order as the momentum. That is $\Delta p \approx p$. It means that sharp position and momentum do not exist simultaneously for the electron in an atom. Hence it is not possible to ascribe any specific trajectory to an electron in an atom. It can only be said that atomic electrons traverse the whole of the space about the nucleus, but however, they move most of the time at a distance corresponding to a permitted Bohr radius.

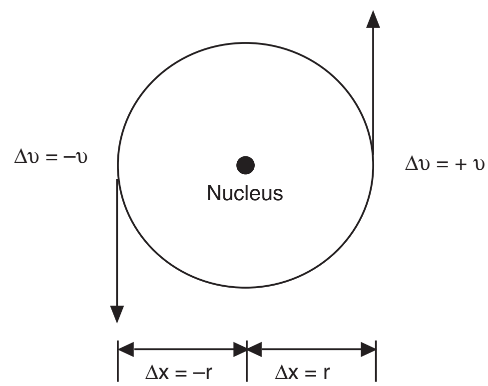

Fig. 20.13: The uncertainties in the position and velocity of electron in an atom.

Now let us calculate the energy of the Bohr's first orbit. The total energy of the electron in the first orbit is given by

$$E = \text{K.E.} + \text{P.E.}$$

$$= \left[\frac{1}{2} m \upsilon^2\right] - \frac{e^2}{4\pi\varepsilon_o r} = \left[\frac{p^2}{2m}\right] - \frac{e^2}{4\pi\varepsilon_o r} \tag{20.39}$$

where p is the momentum of the electron.

As $\Delta p \approx p$, we can write $p \approx \frac{h}{2\pi \Delta x} = \frac{h}{2\pi r}$. (20.40)

$$\therefore \quad E = \frac{h^2}{8\pi^2 m r^2} - \frac{e^2}{4\pi\varepsilon_o r} \tag{20.41}$$

But r is given by $r = \frac{\varepsilon_o h^2}{\pi \, m e^2}$.

$$\therefore \quad E = -\frac{me^4}{8\varepsilon_o^2 h^2} \tag{20.42}$$

The above expression (20.42) is the same as that is given by Bohr theory.

(*b*) Particle in a Box:

Let us consider a particle confined to a box of length l. The uncertainty Δx in the position is l.

$$\Delta x \cdot \Delta p \cong \hbar$$

$$\therefore \quad \Delta p = \frac{\hbar}{\Delta x} = \frac{\hbar}{l} \tag{20.43}$$

Energy is given by
$$E = \frac{p^2}{2m} \approx \frac{(\Delta p)^2}{2m} = \frac{(\hbar / l)^2}{2m} = \frac{\hbar^2}{2ml^2} \quad (20.44)$$

This result agrees with the result obtained from Schrödinger equation. Refer to **§ 20.95.**

(*c*) Electrons cannot be present in the nucleus:

The radiation emitted by radioactive nuclei consists of α, β and γ-rays, out of which β-rays are identified to be electrons. We apply uncertainty principle to find whether electrons are coming out of the nucleus. The radius of the nucleus is of the order of 10^{-14} m. Therefore, if electrons were to be in the nucleus, the maximum uncertainty Δx in the position of the electron is equal to the diameter of the nucleus. Thus,
$$\Delta x = 2 \times 10^{-14} \text{ m.}$$

The minimum uncertainty in its momentum is then given by
$$\Delta p = \frac{\hbar}{\Delta x} = \frac{1.04 \times 10^{-34} J.s}{2 \times 10^{-14} m} = 5.2 \times 10^{-21} \text{ kg-m/s.}$$

The minimum uncertainty in momentum can be taken as the momentum of the electron. Thus,
$$p = 5.2 \times 10^{-21} \text{ kg-m/s.}$$

The minimum energy of the electron in the nucleus is then given by
$$E_{\min} = p_{\min}\, c = (5.2 \times 10^{-21} \text{ kg-m/s})(3 \times 10^8 \text{ m/s}) = 1.56 \times 10^{-12} \text{ J} = 9.7 \text{ MeV.}$$

It implies that if an electron exists within the nucleus, it must have a minimum energy of about 10 MeV. But the experimental measurements showed that the maximum kinetic energies of β-particles were of the order of 4 MeV only. Hence electrons are not present in the nucleus. It is subsequently established that emission of β-particles occurs due to transformations in the nucleus. The transformation of a neutron into a proton produces an electron.

Example 20.4: *Uncertainty in time of an excited atom is about 10^{-8}s. What are the uncertainties in energy and in frequency of the radiation?*

Solution:
$$\Delta E\, \Delta t \approx \frac{h}{2\pi}$$

$$\therefore \quad \Delta E = \frac{1.054 \times 10^{-34} J.s}{10^{-8} s} = \frac{1.054 \times 10^{-26}}{1.602 \times 10^{-19}} eV = 6.58 \times 10^{-8} eV .$$

$$\Delta \nu = \frac{\Delta E}{h} = \frac{1.054 \times 10^{-26} J}{6.626 \times 10^{-34} J.s} = 15.9 \text{ MHz.}$$

Example 20.5. *An electron is confined to a potential well of width 10 nm. Calculate the minimum uncertainty in its velocity.*

Solution.
$$\boldsymbol{\Delta x\, \Delta p} \approx \frac{h}{2\pi} \quad \text{or } \Delta x\, \Delta p \approx \frac{h}{2\pi}$$

$$\therefore \quad \Delta \upsilon = \frac{h}{2\pi\, m \cdot \Delta x}$$

$$\therefore \quad \Delta \upsilon = \frac{6.63 \times 10^{-34} Js}{2 \times 3.143 \times 9.11 \times 10^{-31} kg \times 10 \times 10^{-9} m} = \mathbf{12.1\ km/s}.$$

Example 20.6: *If the kinetic energy of an electron known to be about 1 eV, must be measured to within 0.0001 eV, what accuracy can its position be measured simultaneously?*

Solution: $$E = \frac{p^2}{2m} \quad \therefore \quad \Delta \boldsymbol{E} = \frac{2p\,\Delta p}{2m} \quad \therefore \quad \Delta \boldsymbol{p} = \frac{m}{p}\Delta E$$

$$\Delta x \Delta p = \frac{h}{2\pi}$$

$$\therefore \quad \Delta x = \frac{h}{2\pi \cdot \Delta p} = \frac{h}{2\pi}\cdot\frac{p}{m\Delta E} = \frac{h\sqrt{2mE}}{2\pi\, m\Delta E} = \frac{h}{\pi\,\Delta E}\sqrt{\frac{E}{2m}}$$

$$\therefore \quad \Delta x = \frac{6.63\times10^{-34}\,Js}{3.143\times0.0001\times1.602\times10^{-19}\,J}\cdot\sqrt{\frac{1.602\times10^{-19}\,J}{2\times9.11\times10^{-31}\,kg}}$$

$$= \mathbf{1.95\ \mu m.}$$

Example 20.7: *An electron and a 150 gm base ball are traveling at a velocity of 220 m/s, measured to an accuracy of 0.005 %. Calculate and compare uncertainty in position of each.*

Solution: The uncertainty in the velocity is $\Delta\upsilon = \upsilon\times0.065\% = (220\text{m / s})\times\frac{0.065}{100} = 0.143$ m/s.

(*i*) The uncertainty in the position of electron is

$$\Delta x_e = \frac{\hbar}{2\,m\Delta\upsilon} = \frac{1.05\times10^{-34}\,J.s}{2\times9.11\times10^{-31}\,kg\times0.143\text{ m / s}} = \mathbf{0.4\ mm}.$$

(*ii*) The uncertainty in the position of baseball is

$$\Delta x_B = \frac{\hbar}{2\,M\Delta\upsilon} = \frac{1.05\times10^{-34}\,J.s}{2\times0.15\,kg\times0.143\text{ m / s}} = \mathbf{2.5\times10^{-33}m}.$$

20.17 WAVE FUNCTION AND PROBABILITY INTERPRETATION

Waves represent the propagation of a disturbance in a medium. We are familiar with light waves, sound waves, and water waves. These waves are characterized by some quantity that varies with position and time. Light waves consist of variations of electric and magnetic fields in space, and sound waves consist of pressure variations. We cannot specify in a similar way what is actually varying in de Broglie waves. Since microparticles exhibit wave properties, it is assumed that a quantity ψ represents a de Broglie wave. This quantity ψ is called a **wave function**. ψ describes the wave as a function of position and time. However, it has no direct physical significance, as it is not an *observable* quantity. In general, ψ is a complex-valued function. According to Heisenberg uncertainty principle, we can only know the probable value in a measurement. The probability cannot be negative. Hence ψ cannot be a measure of the presence of the particle at the location (*x, y, z*). But it is certain that it is in someway an index of the presence of the particle at around (*x, y, z, t*).

Probability Interpretation of Wave Function given by Max Born

A probability interpretation of the wave function was given by Max Born in 1926. He suggested that *the square of the magnitude of the wave function* $|\psi|^2$ *evaluated in a particular region represents the probability of finding the particle in that region*. In other words, Probability, P, of finding the particle in an infinitesimal volume dV (= $dx\ dy\ dz$) is proportional to $|\psi(x, y, z)|^2\ dx\ dy\ dz$ at time t.

or $$P \propto |\psi(x,y,z)|^2\ dV \tag{20.45}$$

$|\psi|^2$ is called the **probability density** and ψ is the **probability amplitude**.

Since the particle is certainly somewhere in the space, the probability $P = 1$ and the integral of $|\psi|^2 dV$ over the entire space must be equal to unity. That is

$$\int_{-\infty}^{+\infty} |\psi|^2\, dV = 1 \tag{20.46}$$

The wave function ψ is in general a complex function. But the probability must be real. Therefore to make probability a real quantity, ψ is to be multiplied by its complex conjugate ψ^*.

$$\int_{-\infty}^{+\infty} \psi\psi^*\, dV = 1$$

Thus, ψ has no physical significance but $|\psi|^2$ gives the probability of finding the atomic particle in a particular region.

20.17.1 Normalization Condition

If at all the particle exists, it must be found somewhere in the universe. Since we are sure that the particle must be somewhere in space, the sum of the probabilities over all values of x,y,z must be unity. If $\Psi(x, y, z, t)$ is multiplied by a constant C such that $\Psi_N(x,y,z,t) = C\,\Psi(x,y,z,t)$, where $\Psi_N(x, y, z, t)$ satisfies the relation

$$\int_{-\infty}^{\infty} \left|\Psi_N(x,y,z,t)\right|^2 dxdydz = |C|^2 \int_{-\infty}^{\infty} \left|\Psi(x,y,z,t)\right|^2 dxdydz = 1 \tag{20.47}$$

then $\Psi_N(x,y,z,t)$ is said to be **normalized** wave function and C the **normalization constant.** This condition (20.47) is known as the *normalization condition*. From equ.(20.47), we have

$$|C|^2 = \frac{1}{\displaystyle\int_{-\infty}^{\infty} \left|\Psi(x,y,z,t)\right|^2 dxdydz} \tag{20.48}$$

$|\psi_N(x, y, z, t)|^2\ dxdydz$ is called the **probability density**.

Whenever wave functions are normalised, $|\psi|^2 dV$ equals the probability that a particle will be found in an elemental volume dV.

Thus,

Probability $$P = |\psi(x, y, z)|^2 dV \tag{20.49}$$

20.17.2 Well-behaved Wave Functions - Conditions to be satisfied by ψ-function

An acceptable wave function ψ must be normalized and fulfill the following requirements:

(*i*) ψ **function must be finite:** The wave function must be finite everywhere. Even if $x \to \infty$ or $-\infty$, $y \to \infty$ or $-\infty$, $z \to \infty$ or $-\infty$, the wave function should not tend to infinity. It must remain finite for all values of x, y, z. If ψ is infinite, it would imply an infinitely large probability of finding the particle at that point. This would violate the uncertainty principle.

(*ii*) **ψ function must be single–valued:** Any physical quantity can have only one value at a point. For this reason, the function related to a physical quantity cannot have more than one value at that point. If it has more than one value at a point, it means that there is more than one value of probability of finding the particle at that point (Fig. 20.14).

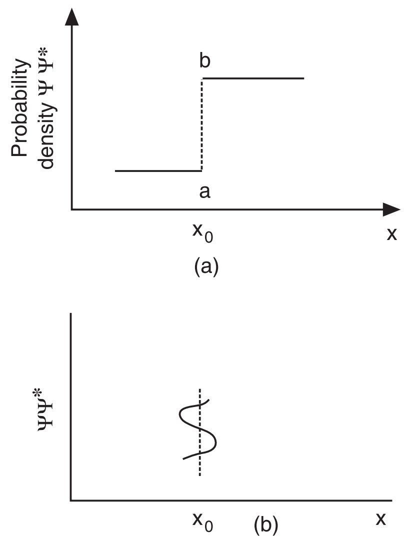

Fig. 20.14

(*iii*) **ψ function must be continuous:** ψ function should be continuous across any boundary. Since ψ is related to a physical quantity, it cannot have a discontinuity at any point. Therefore, the wave function ψ and its space derivatives $\frac{\partial\psi}{\partial x}$, $\frac{\partial\psi}{\partial y}$ and $\frac{\partial\psi}{\partial z}$ should be continuous across any boundary. Since ψ is related to a real particle, it cannot have a discontinuity at any boundary where potential changes.

Wave functions satisfying the above mathematical conditions are called ***well-behaved wave functions.***

20.18 SCHRÖDINGER WAVE EQUATION

In 1926, Erwin Schrödinger, an Austrian physicist, reasoned that the de Broglie waves associated with electrons would resemble the classical waves of light and developed wave equation that describes the behaviour of matter waves. This equation, which has been named after him, defines the wave properties of electrons and also predicts particle-like behaviour.

Time dependent Schrödinger Wave Equation

Since the concept of de Broglie waves is not a result of previous physical theories, it is not possible to derive a wave equation for the wave associated with a particle. However, a wave equation can be developed. Let us assume that a particle of mass '*m*' is in motion along the x-direction. Let the wave function ψ be the dependent variable of the de Broglie wave which is a function of the coordinates x and t. Analogous to the classical wave, we may expect that, ψ will be a function of $(x - \upsilon t)$. As $\upsilon = \omega / k$, the wave function may be written as a function of $(kx - \omega t)$. Using the relation $p = \hbar k$ and $E = \hbar\omega$, we can write

$$\psi = f\left(\frac{px - Et}{\hbar}\right) \tag{20.50}$$

The more general wave would be a sum of a sine and cosine waves. Taking help of Euler's identity, we write the above equation in an exponential form as follows:

$$\psi = A \exp\left[\frac{In}{\hbar}(px - Et)\right] \tag{20.51}$$

We assume that the energy and momentum of the particle are constant. Differentiating the above equation with respect to $x,$ we get

$$\frac{\partial \psi}{\partial x} = A\frac{ip}{\hbar} \ \exp\left[\frac{In}{\hbar}(px - Et)\right]$$

$$\therefore \qquad \frac{\partial \psi}{\partial x} = \frac{ip}{\hbar}\psi$$

Rearranging the terms in the above equation, we get

$$p\psi = \frac{\hbar}{i}\frac{\partial \psi}{\partial x} = -i\hbar\frac{\partial \psi}{\partial x} \tag{20.52}$$

Differentiating the equ.(20.51) with respect to t gives

$$\frac{\partial \psi}{\partial t} = -\frac{iE}{\hbar}A \ \exp\left[\frac{In}{\hbar}(px - Et)\right] = -\frac{iE}{\hbar}\psi$$

Rearranging the terms in the above equation, we get

$$E\psi = -\frac{\hbar}{i}\frac{\partial \psi}{\partial t} = i\hbar\frac{\partial \psi}{\partial t} \tag{20.53}$$

The partial derivatives with respect to x and t are connected by means of the relation between the energy and momentum. The classical expression for the kinetic energy in terms of the momentum is

$$E_k = \frac{m\upsilon^2}{2} = \frac{(m\upsilon)^2}{2m} = \frac{p^2}{2m} \tag{20.54}$$

The total energy and momentum are related by the expression

$$\frac{p^2}{2m} + V = E \tag{20.55}$$

where V is the potential energy of the particle. Multiplying the equ.(20.55) with ψ, we obtain

$$\frac{p^2}{2m}\psi + V\psi = E\psi$$

Using the relations (20.52) and (20.53) into the above equation, we get

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + V\psi = i\hbar\frac{\partial \psi}{\partial t}$$

The above equation may be rewritten as

$$i\hbar\frac{\partial \Psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V\right]\Psi \tag{20.56}$$

The above equation is known as the **time-dependent Schrödinger wave equation.**

When extended to the three-dimensional case, we find that

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2}\right) + V\psi = i\hbar\frac{\partial \psi}{\partial t}$$

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi + V\psi = i\hbar\frac{\partial \psi}{\partial t}$$

or

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = i\hbar\frac{\partial \psi}{\partial t}$$

$$T = Ge^{-2kl} = \left[\frac{16E}{V}\left(1-\frac{E}{V}\right)\right]e^{-2L\sqrt{8\pi^2 m(V-E)}/h}$$

$$\frac{2L}{h}\sqrt{8\pi^2 m\ (V-E)} = \frac{2\times 20\times 10^{-10}m}{6.63\times 10^{-34}Js}\sqrt{8\times(3.143)^2\times 9.11\times 10^{-31}kg(4-3)\times 1.602\times 10^{-19}J}$$

$$= 20.49$$

$$\therefore \qquad T = \left[\frac{16E}{V}\left(1-\frac{E}{V}\right)\right]e^{-20.49}$$

$$= \frac{16\times 3eV}{4eV}\left(1-\frac{3eV}{4eV}\right)e^{-20.49}$$

$$= 3\times e^{-20.49} = 3.8\times 10^{-9}$$

Therefore, the percentage of transmission = $\mathbf{3.8\times 10^{-7}\%}$.

20.22 INFINITE POTENTIAL WELL

A **potential well** is a potential energy function $V(x)$ that has a minimum (see Fig. 20.23). A potential well is the opposite of a potential barrier; it is a potential-energy function with a minimum. If a particle is left in the well and the total energy of the particle is less than the height of the potential well, we say that the particle is *trapped* in the well. In classical mechanics a particle trapped in a potential well can vibrate back and forth with periodic motion but cannot leave the well. In quantum mechanics, such a trapped state is called a **bound state**.

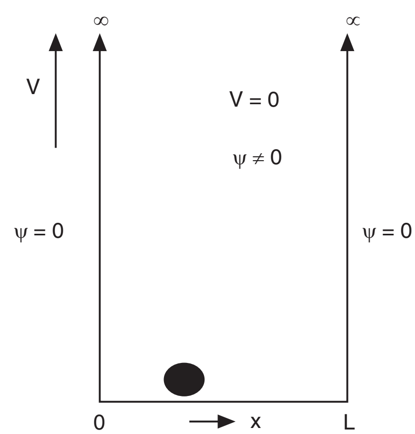

Fig.20.23

Let us consider a particle confined to the region $0 < x < L$. It can move freely within the region $0 < x < L$ but subject to strong forces at $x = 0$ and $x = L$. Therefore, it can never cross to the right to the region $x > L$ or to the left of 0. It means that $V = 0$ in the region $0 < x < L$ and rises to infinity ($V = \infty$) at $x = 0$ and $x = L$. This situation is called a **one-dimensional potential box.**

For a particle trapped in a one-dimensional potential box, $V = 0$ and the Schrodinger equation (20.61a) takes the following form

$$\frac{d^2\psi}{dx^2}+\frac{8\pi^2 mE}{h^2}\psi = 0 \tag{20.90}$$

Putting $\frac{8\pi^2 mE}{h^2} = k^2$, we rewrite the above equation as

$$\frac{d^2\psi}{dx^2}+k^2\psi = 0 \tag{20.91}$$

Because the particle can move back and forth freely between $x = 0$ and $x = \mathrm{L}$, the solution of the equation (20.91) is of the form

$$\psi(x) = Ae^{ikx}+Be^{-ikx} \tag{20.92}$$

which contains motion in both directions. We can evaluate the constants A and B with the help of boundary conditions. The boundary conditions are as follows. The particle cannot jump over the walls and therefore the function does not exist outside the box. The particle is located within the box and therefore it exists within the box. It means that

$$\psi(x) = 0 \text{ at } x = 0$$

$$\psi(x) = 0 \text{ at } x = L$$

$$\therefore \quad \psi_{x=o} = A+B = 0$$

$$\therefore \quad B = -A$$

Using this result into equ.(20.92), we get

$$\psi(x) = A(e^{ikx} - e^{-ikx})$$

or

$$\psi(x) = 2iA \sin kx$$

Again $\psi(x) = 0$ at $x = L$

$$\therefore \quad \psi_{x=L} = 2iA \sin kL = 0$$

The factor $2iA$ cannot be zero, because we would not then have the wave function. It means that

$$\sin kL = 0$$

or

$$kL = n\pi \tag{20.93}$$

As $k = 2\pi/n$, $\lambda = 2L/n$

where n (=1,2,3, ...) is an integer.

The above conclusion implies that the wave equation has solutions only when the particle wavelength is restricted to discrete values such that only a whole number of half-wavelengths are formed over the length L of the box. It means that particle waves form standing wave pattern within the potential box. We write (20.93) as

$$k = n\pi / L$$

(*i*) The possible values of the momentum of the particle are then given by

$$p = \hbar k = \frac{n\pi\hbar}{L} = \frac{nh}{2L} \tag{20.94}$$

(*ii*) The possible values of the energy are

$$E = \frac{p^2}{2m} = \frac{n^2h^2}{8mL^2} \tag{20.95}$$

The above equation indicates that a particle confined in a certain region can have only certain values of energy. Other energy values are not allowed. In other words, **energy quantization** is a consequence of restricting a microparticle to a certain region.

The allowed wave functions which are the solutions of the Schrodinger equation are given by

$$\psi_n = 2iA \sin kx = C \sin kx$$

As $k = n\pi / L$, we write the above as

$$\psi_n = C \sin\left(\frac{n\pi x}{L}\right) \tag{20.96}$$

Applying normalization condition, we can determine the value of the constant C in the above equation. The normalisation condition is that